

PRAKTIKUM BASIS DATA

MODUL 13

NoSQL – Document Database



Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

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MODUL 13

TUGAS JURNAL

OUTPUT: SCREENSHOOT LAYAR HASIL RUNNING QUERY MONGOSHELL

Disediakan dataset berikut:

```
db.toko.insertMany([
  {
    "name"      :
    "Toko Berkah",
    "category" :
    "Elektronik", "city" :
    "Jakarta",
    "address" : { "street" : "Jl. Sudirman", "zipcode" : "10220" },
    "products" : [
      { "item" : "TV",    "price" : 3500000, "stock" : 10 },
      { "item" : "Kulkas", "price" : 2800000, "stock" : 5 }
    ]
  },
  {
    "name"      :
    "Toko Maju", "category"
    : "Elektronik", "city":
    "Bandung",
    "address" : { "street" : "Jl. Asia Afrika", "zipcode" : "40111" },
    "products" : [
      { "item" : "HP",    "price" : 1500000, "stock" : 20 },
      { "item" : "Laptop", "price" : 8000000, "stock" : 3 }
    ]
  },
  {
    "name"      : "Toko Salim
    Bakery", "category" :
    "Elektronik",
    "city"      : "Jakarta",
    "address" : { "street" : "Jl. Sudirman", "zipcode" : "10220" },
    "products" : [
      { "item" : "Monitor LED",  "price" : 7700000, "stock" : 11 },
      { "item" : "Airfryer Panasonic", "price" : 4850000, "stock" : 7 }
    ]
  },
  {
    "name"      : "Toko
```

```

    { "item" : "RTX 55000", "price" : 8500000, "stock" : 20 },
    { "item" : "Set PC Gaming Radeon", "price" : 12000000, "stock" : 9 }
  ]
},
))

```

- 1.) Tambahkan 1 toko baru bebas ke koleksi toko, dengan ketentuan:
 - Minimal memiliki 2 produk di dalam array products
 - Isi semua field: name, category, city, address, products

Hint:

```

db.toko.insertOne({
  "name" : "...",
  "category" : "...",
  "city" : "...",
  "address" : { "street" : "...", "zipcode" : "..." }, "products"
  : [
    { "item" : "...", "price" : ..., "stock" : ... },
    { "item" : "...", "price" : ..., "stock" : ... }
  ]
})

```

- 2.) Tampilkan semua toko yang berada di kota Jakarta.
- 3.) Tampilkan toko yang memiliki produk dengan price lebih dari 3.000.000.
Hint:
 - Gunakan dot notation: products.price
 - Gunakan operator: \$gt
- 4.) Ubah category Toko Berkah dari Fashion menjadi Aksesoris.
Tambahkan 1 produk baru ke dalam array products milik Toko Maju menggunakan \$push.

Hint:

- Filter berdasarkan name: Toko Maju
 - Gunakan operator \$push untuk menambah elemen ke array
- 5.) Hitung jumlah toko per city, urutkan dari kota yang paling banyak tokonya.

Gunakan:

- \$group dengan \$sum: 1
- \$sort descending (-1)

- 6.) Hitung rata-rata harga produk per category.

Tahapan pipeline:

- \$unwind: '\$products' (bongkar array products dulu)
- \$group: kelompokkan per category, hitung \$avg dari products.price
- \$sort: urutkan sesuai kebutuhan

7.) Tambahkan 1 produk baru ke dalam array products milik Toko Maju menggunakan \$push.

Hint:

- Filter berdasarkan name: Toko Maju
- Gunakan operator \$push untuk menambah elemen ke array

8.) Tampilkan harga produk tertinggi (\$max) per city.

Hint:

- Unwind products terlebih dahulu
- Gunakan \$max pada products.price

9.) Buat pipeline lengkap dengan 4 tahap:

- \$match — filter hanya toko category Elektronik
- \$unwind — bongkar array products
- \$group — hitung total stock per city
- \$sort — urutkan dari total stock terbesar

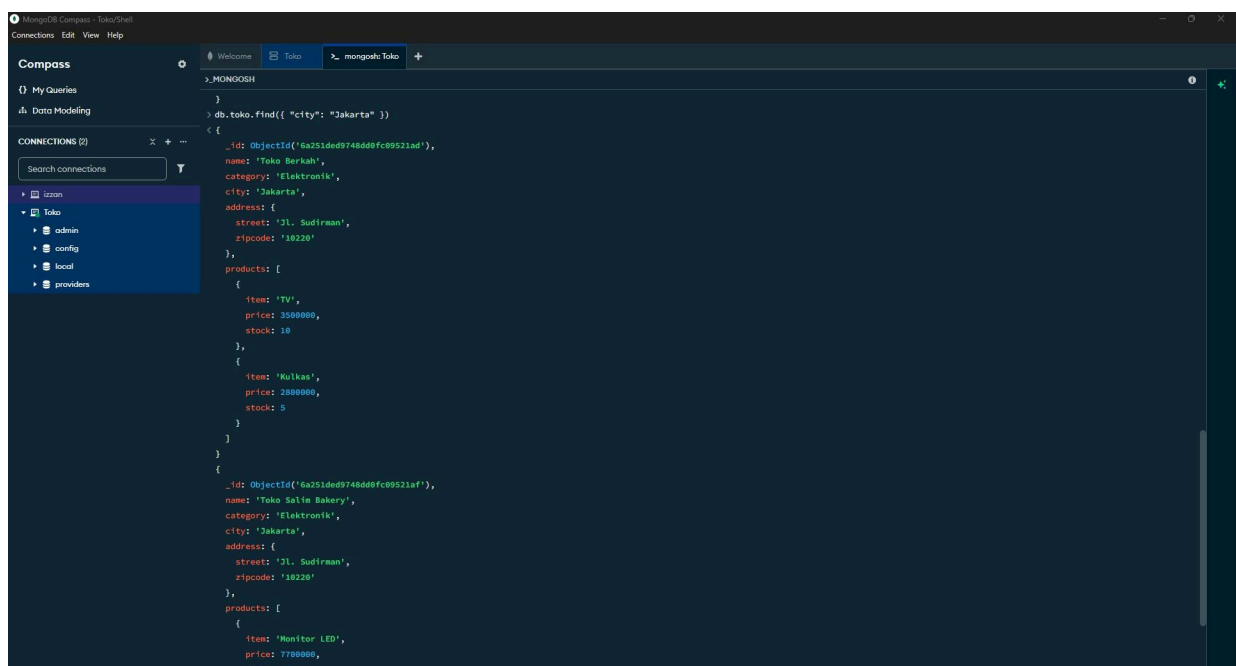
10.) Hapus toko yang memenuhi dua kondisi sekaligus:

- category adalah Fashion
- city adalah Surabaya

Hint:

- Gunakan \$and atau tulis kedua kondisi dalam satu objek filter

Jawaban:



```
MongoDB Compass - Toko/Shell
Connections Edit View Help

Compass
() My Queries
Data Modeling

CONNECTIONS (2)
Search connections
+ izon
+ Toko
  + admin
  + config
  + local
  + providers

Welcome Toko mongo: Toko +
> MONGODB
> db.toko.find({ "products.price": { $gt: 3000000 } })
<
[
  {
    _id: ObjectId('6a251ded9748dd0fc09521ad'),
    name: 'Toko Berkah',
    category: 'Elektronik',
    city: 'Jakarta',
    address: {
      street: 'Jl. Sudirman',
      zipcode: '10220'
    },
    products: [
      {
        item: 'TV',
        price: 3500000,
        stock: 10
      },
      {
        item: 'Kulkas',
        price: 2800000,
        stock: 5
      }
    ]
  },
  {
    _id: ObjectId('6a251ded9748dd0fc09521ae'),
    name: 'Toko Maju',
    category: 'Elektronik',
    city: 'Bandung',
    address: {
      street: 'Jl. Asia Afrika',
      zipcode: '40111'
    },
    products: [
      {
        item: 'HP',
        price: 3500000,
        stock: 20
      }
    ]
  }
]
```

```
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CONNECTIONS (2)
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Welcome Toko mongo: Toko +
> MONGODB
    stock: 7
  }
}
{
  _id: ObjectId('6a251ded9748dd0fc09521be'),
  name: 'Toko Sudinotech',
  category: 'Elektronik',
  city: 'Surabaya',
  address: {
    street: 'Jl. Sudirman',
    zipcode: '10220'
  },
  products: [
    {
      item: 'RTX 55000',
      price: 8500000,
      stock: 20
    },
    {
      item: 'Set PC Gaming Radeon',
      price: 12000000,
      stock: 9
    }
  ]
}
> db.toko.updateOne(
  { "name": "Toko Berkah" },
  { $set: { "category": "Aksesoris" } }
)
<
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
toko>
```

```
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  + providers

Welcome Toko mongo: Toko +
    item: 'Set PC Gaming Radeon',
    price: 12000000,
    stock: 9
  }
}
> db.toko.updateOne(
  { "name": "Toko Berkah" },
  { $set: { "category": "Aksesoris" } }
)
<
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
> db.toko.aggregate([
  { $group: { _id: "city", jumlahToko: { $sum: 1 } } },
  { $sort: { jumlahToko: -1 } }
])
<
[
  {
    _id: 'Jakarta',
    jumlahToko: 2
  },
  {
    _id: 'Bandung',
    jumlahToko: 1
  },
  {
    _id: 'Surabaya',
    jumlahToko: 1
  },
  {
    _id: 'Semarang',
    jumlahToko: 1
  }
]
toko>
```

```
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CONNECTIONS (2)
Search connections
+ izoon
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+ providers

Welcome Toko mongosh Toko +
> use TOKO
> db.toko.aggregate([
  { $group: { _id: "$city", jumlahToko: { $sum: 1 } } },
  { $sort: { jumlahToko: -1 } }
])
< {
  _id: 'Jakarta',
  jumlahToko: 2
}
{
  _id: 'Bandung',
  jumlahToko: 1
}
{
  _id: 'Surabaya',
  jumlahToko: 1
}
{
  _id: 'Semarang',
  jumlahToko: 1
}
]
> db.toko.aggregate([
  { $unwind: "$products" },
  { $group: { _id: "category", rataHarga: { $avg: "$products.price" } } }
])
< {
  _id: 'Pakaian',
  rataHarga: 200000
}
{
  _id: 'Aksesoris',
  rataHarga: 3150000
}
{
  _id: 'Elektronik',
  rataHarga: 7091666.666666667
}
}
>
> toko>
```

```
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> use TOKO
> db.toko.aggregate([
  { $unwind: "$products" },
  { $group: { _id: "category", rataHarga: { $avg: "$products.price" } } }
])
< {
  _id: 'Pakaian',
  rataHarga: 200000
}
{
  _id: 'Aksesoris',
  rataHarga: 3150000
}
{
  _id: 'Elektronik',
  rataHarga: 7091666.666666667
}
}
> db.toko.updateOne(
  { "name": "Toko Maju" },
  { $push: { "products": { "item": "Mouse Wireless", "price": 150000, "stock": 15 } } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
>
> toko>
```

```
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Welcome Toko mongosh Toko +
> use TOKO
> db.toko.updateOne(
  { "name": "Toko Maju" },
  { $push: { "products": { "item": "Mouse Wireless", "price": 150000, "stock": 15 } } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
> db.toko.aggregate([
  { $unwind: "$products" },
  { $group: { _id: "city", hargaTertinggi: { $max: "$products.price" } } }
])
< {
  _id: 'Surabaya',
  hargaTertinggi: 12000000
}
{
  _id: 'Jakarta',
  hargaTertinggi: 7700000
}
{
  _id: 'Bandung',
  hargaTertinggi: 8000000
}
{
  _id: 'Semarang',
  hargaTertinggi: 250000
}
}
>
> toko>
```

```
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Welcome Toko mongo: toko
> use toko
switched to db toko

{ $unwind: "$products" },
{ $group: { _id: "$city", hargaTertinggi: { $max: "$products.price" } } }
})

< {
  _id: 'Surabaya',
  hargaTertinggi: 12000000
}
{
  _id: 'Jakarta',
  hargaTertinggi: 7700000
}
{
  _id: 'Bandung',
  hargaTertinggi: 8800000
}
{
  _id: 'Semarang',
  hargaTertinggi: 250000
}
}

> db.toko.aggregate([
  { $match: { "category": "Elektronik" } },
  { $unwind: "$products" },
  { $group: { _id: "$city", totalStock: { $sum: "$products.stock" } } },
  { $sort: { totalStock: -1 } }
])

< {
  _id: 'Bandung',
  totalStock: 38
}
{
  _id: 'Surabaya',
  totalStock: 29
}
{
  _id: 'Jakarta',
  totalStock: 18
}
}

toko>
```

```
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Welcome Toko mongo: toko
> use toko
switched to db toko

{
  _id: 'Bandung',
  hargaTertinggi: 8800000
}
{
  _id: 'Semarang',
  hargaTertinggi: 250000
}
}

> db.toko.aggregate([
  { $match: { "category": "Elektronik" } },
  { $unwind: "$products" },
  { $group: { _id: "$city", totalStock: { $sum: "$products.stock" } } },
  { $sort: { totalStock: -1 } }
])

< {
  _id: 'Bandung',
  totalStock: 38
}
{
  _id: 'Surabaya',
  totalStock: 29
}
{
  _id: 'Jakarta',
  totalStock: 18
}
}

> db.toko.deleteMany({
  $and: [
    { "category": "Fashion" },
    { "city": "Surabaya" }
  ]
})

< {
  acknowledged: true,
  deletedCount: 0
}

toko>
```